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CASE STUDY

AI Communication Orchestration Platform

Backend Engineering / AI Systems / Workflow Automation

OVERVIEW

I built a platform that turns natural-language requests into coordinated communication workflows across email, calendar, WhatsApp, SMS, and voice. It combines conversational AI, durable orchestration, and persistent memory while keeping sensitive actions subject to human review.

CHALLENGE

Business communication is spread across inboxes, messaging tools, and calendars. Bringing these channels into one conversational interface required reliable execution, clear permissions, and continuity between interactions. Incorrect recipients, unsupported confirmations, and external API failures all had to be addressed in the architecture.

MY ROLE

I designed and implemented the backend architecture, Temporal workflows, AI integration, approval mechanisms, contact and conversation memory, and communication integrations. My work covered both multi-turn conversations and compound workflows spanning external services.

CORE DECISION

Separate conversation from execution. AI interprets requests and proposes next steps; a controlled workflow validates inputs, enforces approval requirements, and carries out external actions.

SYSTEM DESIGN

Architecture and execution

A controlled path from incoming message to external action

GATEWAY

FastAPI receives and authenticates webhook events, normalizes payloads, and starts workflows. The ingress layer holds no business logic, AI reasoning, or persistent state.

ORCHESTRATION

Temporal coordinates workflow state, activities, retries, and recovery. ConversationWorkflow manages context, gathers missing information, and generates responses. IntentWorkflow validates executable requests and coordinates external actions.

AI AND MEMORY

Azure-hosted models support intent extraction, classification, message generation, and call summarization. Generated outputs are validated before use. PostgreSQL stores contacts, channel identities, conversations, interaction history, and pending actions to preserve context across channels.

ACTION PATH

Incoming event > Gateway > ConversationWorkflow
Action request > Authority and approval checks
Approved request > IntentWorkflow > External service

Conversation can also produce a response or retrieve information without executing a business action.

HUMAN CONTROL

Sensitive decisions are escalated for review. The conversation layer cannot execute business actions directly; external actions pass through the execution workflow. Temporal records execution history for traceability and recovery.

DELIVERY

Capabilities and outcomes

A working platform for communication and operational workflows

DELIVERED	Multi-channel conversations; email and calendar automation; WhatsApp, SMS, and voice support; compound workflows; human approval and escalation; contact resolution; and persistent conversation memory.
RELIABILITY	Durable workflows coordinate asynchronous work and retries when external services fail. Separating AI-generated plans from controlled execution gives operators a clearer review path and an execution history they can inspect.
TRADEOFF	Direct model-to-service execution would simplify the system and reduce latency. I chose a layered workflow architecture to gain explicit validation, approval pathways, and traceability. The tradeoff is greater implementation and orchestration complexity.
TECHNOLOGY	Backend: Python, FastAPI, Uvicorn, Temporal, Docker Data: PostgreSQL, SQLAlchemy, Alembic AI: Azure OpenAI, GPT-4.1 Mini, GPT-4o Channels: Gmail API, Google Calendar API, Meta WhatsApp Cloud API, Twilio Voice: Pipecat, Deepgram, Cartesia
TAKEAWAY	The project demonstrates an approach to using AI in operational software: natural-language interaction supported by durable workflows, persistent context, and explicit human control. The delivered outcome is a working platform that unifies communication and operational workflows.